Train a deep model to classify lung diseases

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Abstract

Large-scale chest X-ray classification is investigated using a deep learning architecture, ConvNeXt, trained on the ChestX-ray14 dataset. All images are resized to 224×224 *pixels, and the task is formulated as multi-label classification over 14 thoracic disease categories with no finding label. Model performance is evaluated using micro- and macro-averaged area under the ROC curve (AUC). To ensure robustness, each experiment is repeated ten times, and the mean and standard deviation of AUC are reported.*

Experimental results indicate that directly training ConvNeXt fine-tuned from ImageNet cwith binary cross-entropy with logits and adopting sigmoid-based probability estimation substantially improves convergence, achieving 0.83 AUC and lower variance, particularly for rare disease categories. These findings highlight the critical role of loss design and transfer learning in large-scale medical image classification.

1. Introduction

In the medical domain, large volumes of imaging data require expert interpretation to identify subtle or latent pathological patterns. Manual diagnosis is time-consuming and may lead to diagnostic errors, particularly false negatives. To address these challenges, machine learning techniques have been increasingly adopted to assist clinicians by improving diagnostic accuracy, reducing false negative rates, and increasing true positive detection. Training robust models on large and diverse datasets is critical for enabling the discovery of disease patterns that may be difficult to identify through manual inspection.

This study investigates a ConvNeXt model, for large-scale chest X-ray classification on the ChestX-ray14 dataset. All images are preprocessed to a uniform resolution of 224×224 pixels and converted to single-channel grayscale format. The task is formulated as multi-label classification to account for the presence of multiple co-occurring thoracic diseases.

2. Dataset

The ChestX-ray14 dataset contains 112,120 frontal chest radiographs annotated with 14 thoracic disease labels and an additional No Finding label. Following common practice, 25,596 images are reserved for testing, while the remaining images are split into 75,312 training images and 11,212 validation images.

All images are resized to 224×224 pixels and normalized using ImageNet statistics. During training, data augmentation is applied in the form of random horizontal flipping and small random rotations. No augmentation is applied during validation or testing. All splits follow the official patient-independent partitioning to avoid data leakage.

3. ConvNeXt Architecture

ConvNeXt is adopted as the backbone network for chest radiograph analysis. ConvNeXt is a modernized convolutional architecture that incorporates design principles inspired by Vision Transformers while retaining the inductive biases of convolutional neural networks. The architecture follows a hierarchical design composed of multiple stages with progressively increasing channel dimensions and decreasing spatial resolutions.

Each stage consists of repeated ConvNeXt blocks. A ConvNeXt block contains a depthwise convolution with a large kernel size (7×7) to efficiently enlarge the receptive field, followed by Layer Normalization applied in a channels-last format. The normalized features are passed through a pointwise expansion layer, a GELU non-linearity, and a pointwise projection layer. Residual connections are employed to facilitate gradient flow and improve optimization stability.

The model is optimized using binary cross-entropy loss with logits and the AdamW optimizer, with an initial learning rate of 3×10^{-4} and a weight decay of 0.05. A batch size of 32 is used, and training is conducted for a fixed number of epochs. The checkpoint with the lowest validation loss is selected for evaluation.

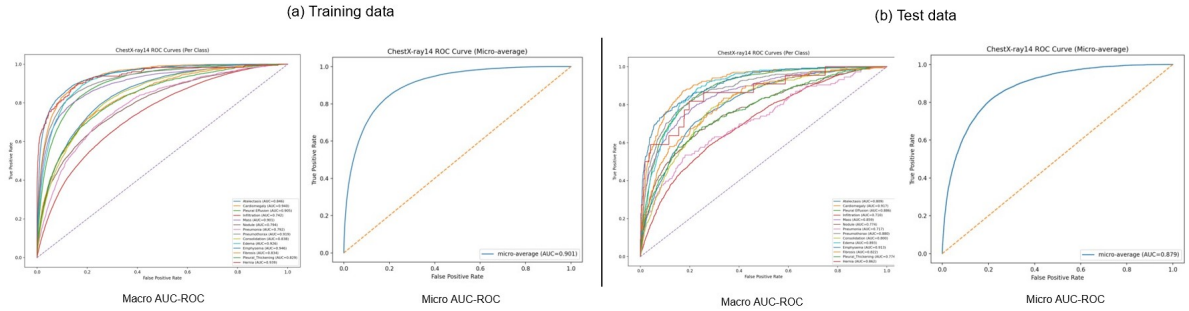

Figure 1. Training and test ROC curves for ImageNet pretraining, showing micro- and macro-averaged AUC at the best epoch.

4. Experiment

The classification task is formulated as a multi-label problem with $C = 15$ disease categories. Given an input image x, the ConvNeXt backbone produces a feature representation $f(x)$. The original ImageNet classifier is replaced with a linear layer that outputs C logits:

$$\mathbf{z} = Wf(x) + b, \quad \mathbf{z} \in \mathbb{R}^{14}. \tag{1}$$

Independent sigmoid activations are applied to obtain class-wise probabilities:

$$\hat{\mathbf{y}} = \sigma(\mathbf{z}). \tag{2}$$

Model performance is evaluated using both micro- and macro-averaged AUC. Micro-AUC aggregates predictions across all disease categories and reflects overall performance, while macro-AUC averages per-class AUCs and evaluates balanced performance across diseases.

Fine-tuning from ImageNet: ConvNeXt is initialized with ImageNet-pretrained weights and fine-tuned on the ChestX-ray14 training set. The best-performing checkpoint is obtained at epoch 7 based on validation loss.

As shown in Figure 1, ImageNet-pretrained models achieve higher ROC-AUC values across both training and test sets, with micro-AUC values approaching 0.90. The relatively small gap between training and test performance indicates good generalization and limited overfitting.

Pretraining	**Micro AUC**	**Macro AUC**
ImageNet	0.901 ± 0.0098	0.866 ± 0.0063

Table 1. Comparison of micro- and macro-averaged AUC for models trained on training data from scratch and with ImageNet pretraining.

Pretraining	**Micro AUC**	**Macro AUC**
ImageNet	0.879 ± 0.0145	0.830 ± 0.0124

Table 2. Comparison of micro- and macro-averaged AUC for models trained on test data from scratch and with ImageNet pretraining.

4.1. Summary

Experimental results indicate that Swin-B is highly sensitive to the choice of loss formulation and probability estimation. Employing binary cross-entropy with logits and sigmoid-based predictions is essential for achieving meaningful performance in multi-label chest X-ray classification.

ConvNeXt fine-tuned from ImageNet consistently outperforms training from scratch, achieving higher AUC and lower variance across both common and rare thoracic diseases. These findings underscore the importance of transfer learning for medical image analysis, even in the presence of large-scale labeled datasets.

Furthermore, training a single backbone across multiple chest X-ray datasets can be effectively achieved by aligning heterogeneous annotations into a unified label space and applying masked multi-label supervision.